Saee Desai

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Summary

M.S. Data Science graduate building **Python AI/ML applications** across **agentic RAG, transformer fine-tuning, model evaluation, and reliable inference**. Developed a deployed **AWS/FastAPI inference gateway** with resilience and observability, a **LangGraph research assistant**, and an NLP moderation prototype. Two first-author publications in post-quantum security and applied cryptography.

Education

University of Maryland, College Park — Maryland, USA
***M.S. in Data Science**, GPA: 3.8/4.0* — *Aug 2024 – May 2026*

Vishwakarma Institute of Information Technology — Pune, India
***B.Tech in Computer Engineering**, CGPA: 9.13/10* — *Aug 2020 – May 2024*

Research Experience

Applied Researcher — Sep 2025 – May 2026
Do Quantum, UMD student research organization — *College Park, Maryland*

- Consulted a pharmaceutical R&D digital-transformation expert on adverse-event workflows and privacy needs. Built a **Python/Flask research prototype** simulating hospital-to-pharma data exchange with **ML-KEM-768, ML-DSA-65, HKDF-SHA-256, and AES-256-GCM**.
- Tested **ciphertext tampering, corruption, truncation, and replay-style modifications**, examining signature/decryption failures and documenting limitations, including unsigned transmission metadata.
- Benchmarked synthetic **TXT, CSV, JSON, and PDF payloads (1–10 MB)**; found processing costs increased with payload size while key-establishment overhead remained relatively constant, contributing to first-author research.

Technical Skills

Languages & Data: Python, SQL, C++, Pandas, NumPy, Statistical Analysis, Data Preprocessing
AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, DistilBERT, Classification, Fine-Tuning, Model Evaluation, Error Analysis
LLMs & Retrieval: LangGraph, Agentic RAG, FAISS, Sentence-Transformer Embeddings, Semantic Search, Tool Routing, Retrieval Evaluation, LLM-as-Judge
Backend & Reliability: FastAPI, Flask, REST APIs, Async Python, Redis, pytest, Retries, Circuit Breakers, Failover, Rate Limiting, Backpressure, Locust
Cloud & DevOps: AWS (ECS Fargate, ALB, ECR, SSM, S3), Docker, Terraform, GitHub Actions CI/CD, OIDC, Prometheus, Grafana, Git, Gradio

Projects

Inference Reliability Gateway | GitHub — June 2026 – July 2026

- Built an **async FastAPI LLM inference gateway** with **Redis caching**, complexity-based model routing, and a shared interface across inference backends.
- Implemented **timeouts, retries, circuit breakers, failover, rate limits, and bounded-concurrency backpressure**; validated behaviors through **15 pytest tests** and controlled failure scenarios.
- Instrumented **Prometheus and Grafana**, load-tested with Locust, and measured **624 ms uncached vs. 5.5 ms cached** in one repeat-request test; estimated **42% lower inference cost** on a 12-prompt routing comparison.
- Deployed with **Docker, Terraform, AWS ECS Fargate, ALB, ECR, and SSM**; configured **GitHub Actions CI/CD via OIDC** and tested autoscaling from **1 to 4 tasks**.

Agentic RAG Research Assistant | GitHub | Demo — May 2026 – June 2026

- Ingested **20 research PDFs from AWS S3**, extracted text into **1,953 overlapping chunks**, and indexed **all-MiniLM-L6-v2** embeddings in **FAISS** for semantic retrieval.
- Developed a **LangGraph LLM planner** routing queries to paper retrieval, **Tavily** web search, both, or neither; integrated **Groq** generation and deployed a **Gradio** interface.
- Evaluated retrieval and answer quality on a **six-case pilot**, using **LLM-as-Judge** ratings for faithfulness and relevance alongside retrieval metrics.
- Identified degraded results after adding a cross-encoder reranker, then removed the reranker and applied metadata filtering to address observed retrieval failures.

Reddit Hate Speech Detection & Emotion Analysis | GitHub — NLP Course Project

- Co-developed two task-specific **fine-tuned DistilBERT classifiers** for binary hate/toxicity detection on **HateXplain** and coarse-grained emotion prediction on **GoEmotions**.
- Evaluated **76.25% hate-detection accuracy** and **0.735 macro-F1** for emotion classification; compared pretrained transformer baselines and examined confusion-matrix error patterns.
- Integrated a lightweight **inference API** with Reddit post/comment ingestion and confidence-scored predictions for **human-review-oriented** thread exploration.

Publications

- **Towards Post-Quantum Secure Pharmacovigilance with ML-KEM and ML-DSA** *(First Author). arXiv preprint*, arXiv:2606.09412 [cs.CR], 2026.
- **Enhanced Communication Security: Leveraging Quantum Cryptography** *(First Author). ICT4SD 2024*, Lecture Notes in Networks and Systems, vol. 1197. Springer, 2025.